# Ceiling of a Number (medium)

## Problem Statement

Given an array of numbers sorted in an ascending order, find the ceiling of a given number ‘key’. The ceiling of the ‘key’ will be the smallest element in the given array greater than or equal to the ‘key’.

Write a function to return the index of the ceiling of the ‘key’. If there isn’t any ceiling return -1.

**Example 1:**

```
Input: [4, 6, 10], key = 6
Output: 1
Explanation: The smallest number greater than or equal to '6' is '6'
having index '1'.
```

**Example 2:**

```
Input: [1, 3, 8, 10, 15], key = 12
Output: 4
Explanation: The smallest number greater than or equal to '12' is '15'
having index '4'.
```

**Example 3:**

```
Input: [4, 6, 10], key = 17
Output: -1
Explanation: There is no number greater than or equal to '17' in the
given array.
```

**Example 4:**

```
Input: [4, 6, 10], key = -1
Output: 0
Explanation: The smallest number greater than or equal to '-1' is '4'
having index '0'.
```

## Try it yourself

Try solving this question here:

Java

```java
class Solution {

  public static int searchCeilingOfANumber(int[] arr, int key) {
    // TODO: Write your code here
    return -1;
  }
}
```

Test Case | Run Code Result | Stdout

```
[4,6,10]
10
```

Use Example Testcases | Run | Submit

← Previous
Solution: Order-Agnostic Binary Search

Next →
Solution: Ceiling Of A Number

Mark as Completed